CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 16: Understanding Shapes

Topic-wise short notes with solved examples

Curves and polygons, convex and concave polygons, regular polygons, the sum of the interior and exterior angles of a polygon, and the number of diagonals.

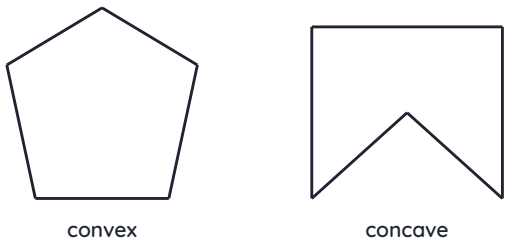
1 Curves and polygons

- A **simple curve** does not cross itself. A **closed curve** begins and ends at the same point.
- A **polygon** is a simple closed curve made only of line segments. The segments are its **sides** and their end points its **vertices**.
- A **diagonal** joins two vertices that are not next to each other.
- **Adjacent sides** share a vertex. **Adjacent vertices** are the ends of one side.

Number of sides	3	4	5	6	7	8	9	10
Name	Triangle	Quadrilateral	Pentagon	Hexagon	Heptagon	Octagon	Nonagon	Decagon

2 Convex and concave polygons

- **Convex polygon**: every interior angle is less than 180°. All its diagonals lie inside it.
- **Concave polygon**: at least one interior angle is greater than 180° (a reflex angle). At least one diagonal lies partly outside.
- Unless told otherwise, "polygon" means a convex polygon.



A convex pentagon and a concave pentagon

3 Regular polygons

- A **regular polygon** has **all sides equal** and **all angles equal**. Examples: equilateral triangle, square, regular hexagon.
- A rhombus has equal sides but is not regular (its angles differ). A rectangle has equal angles but is not regular (its sides differ).

4 Interior angles

- A polygon of n sides can be divided into $n - 2$ triangles by the diagonals from one vertex.

$$\text{Sum of the interior angles} = (n-2) \times 180^\circ = (2n-4) \text{ right angles}$$

Polygon	Sides	Sum of interior angles	Each angle if regular
Triangle	3	180°	60°
Quadrilateral	4	360°	90°
Pentagon	5	540°	108°
Hexagon	6	720°	120°
Octagon	8	1080°	135°
Decagon	10	1440°	144°

- Each interior angle of a regular polygon $= \dfrac{(n-2) \times 180^\circ}{n}$.

Example 4.1 Find the sum of the interior angles of a polygon with 12 sides.

SOLUTION

$(12-2) \times 180^\circ$

1800°

Example 4.2 The sum of the interior angles of a polygon is 1260°. How many sides does it have?

SOLUTION

$(n-2) \times 180 = 1260$, so $n - 2 = 7$

9 sides

Example 4.3 Four angles of a pentagon are 100°, 110°, 95° and 125°. Find the fifth.

SOLUTION

$540^\circ - 430^\circ$

110°

Example 4.4 The angles of a hexagon are x, $x + 10^\circ$, $x + 20^\circ$, $x + 30^\circ$, $x + 40^\circ$ and $x + 50^\circ$. Find x.

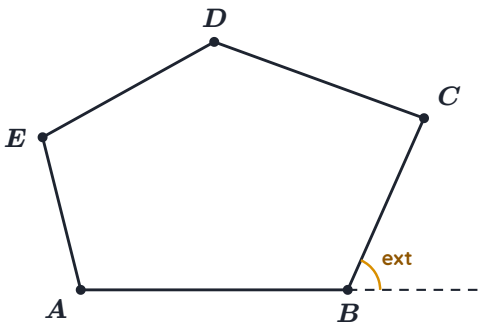
SOLUTION

$6x + 150 = 720$

$x = 95^\circ$

5 Exterior angles

- An **exterior angle** is formed when a side is produced. At each vertex, interior angle + exterior angle $= 180^\circ$.
- **The sum of the exterior angles of any convex polygon, one at each vertex, is 360°.**
- For a regular polygon of n sides: each exterior angle $= \dfrac{360^\circ}{n}$, so $n = \dfrac{360^\circ}{\text{exterior angle}}$.
- Each interior angle of a regular polygon $= 180^\circ - \dfrac{360^\circ}{n}$.



The exterior angle at B is formed by producing AB

Example 5.1 Find each exterior angle and each interior angle of a regular polygon of 15 sides.

SOLUTION

Exterior: $360^\circ \div 15$. Interior: $180^\circ - 24^\circ$.

24° **and** 156°

Example 5.2 Each exterior angle of a regular polygon is 40°. How many sides does it have?

SOLUTION

$360 \div 40$

9

Example 5.3 Each interior angle of a regular polygon is 140°. Find the number of sides.

SOLUTION

Exterior angle $= 40^\circ$

9

Example 5.4 Can a regular polygon have each exterior angle equal to 50°?

SOLUTION

$360 \div 50 = 7.2$, which is not a whole number.

No

Example 5.5 The ratio of each interior angle to each exterior angle of a regular polygon is $4 : 1$. Find the number of sides.

SOLUTION

$4x + x = 180^\circ$, so the exterior angle is 36°.

10 sides

6 Diagonals

- From one vertex of an n-sided polygon, $n - 3$ diagonals can be drawn.

$$\text{Number of diagonals} = \frac{n(n-3)}{2}$$

Example 6.1 How many diagonals does an octagon have?

SOLUTION

$$\frac{8 \times 5}{2}$$

20

Example 6.2 A polygon has 35 diagonals. How many sides does it have?

SOLUTION

$n(n-3) = 70 = 10 \times 7$

10 sides

7 Common mistakes

- Using $(n-2) \times 180^\circ$ for the sum of the exterior angles. That sum is always 360°.
- Taking $n \times 180^\circ$ as the interior angle sum.
- Calling a rhombus or a rectangle a regular polygon.
- Finding the number of sides from the interior angle by dividing 360° by it. Use the exterior angle.
- Counting sides as diagonals.
- Accepting a number of sides that is not a whole number.

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